Solving Trigonometric Equations on a Restricted Interval

Solving linear, factorable, identity-first and multiple-angle trigonometric equations on a restricted interval

Directions.

- Give *every* solution inside the stated interval, and no solution outside it. List them in increasing order.
- Angles are in degrees wherever a degree symbol appears; problem 6 is in hours.
- For a multiple-angle equation, widen the interval for the inside angle first, solve there, then divide.

1. Solve $2\cos\theta + 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.

Answer: ________________ degrees

2. Solve $\sqrt{2}\sin\theta - 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.

Answer: ________________ degrees

3. Solve $\sin\theta\,(2\sin\theta - 1) = 0$ for $0^\circ \le \theta < 360^\circ$. Give all four solutions.

Answer: ________________ degrees

4. A cam on a press returns to its marked position whenever the crank angle θ satisfies $2\sin 2\theta - \sqrt{3} = 0$, with θ measured in degrees. Find every crank angle in one full turn, $0^\circ \le \theta < 360^\circ$, at which this happens.

Answer: ________________ degrees

5. Consider $2\sin^2\theta + 3\cos\theta - 3 = 0$ on $0^\circ \leq \theta < 360^\circ$.

(a) Use a Pythagorean identity to rewrite the left-hand side as a product of two factors.

Answer: ________________

(b) Solve the equation on the given interval.

Answer: ________________ degrees

6. A buoy rides a swell so that its height above the mean water line is $h(t) = 6\sin\left(\dfrac{\pi t}{6}\right)$, where h is in metres and t is the number of hours after midnight.

(a) Find every time t with $0 \leq t < 12$ at which the buoy is exactly 3 metres above the mean line.

Answer: ________________ hours

(b) Explain what those two times mean for the buoy's motion during those twelve hours.

Answer: ________________

7. Solve $2\cos 3\theta - 1 = 0$ for $0° \leq \theta < 180°$.

Answer: ________________ degrees

8. Consider $2\cos^2\theta - \sin\theta - 1 = 0$ on $0° \leq \theta < 360°$.

(a) Use a Pythagorean identity to rewrite the left-hand side as a product of two factors.

Answer: ________________

(b) Solve the equation on the given interval.

Answer: ________________ degrees

9. A searchlight sweeps a wall. The brightness at a fixed point on the wall is $B(\theta) = 40 \sin 2\theta$ lux, where θ is the beam angle in degrees. Find every beam angle with $0° \le \theta < 180°$ at which the brightness is exactly 20 lux.

Answer: ________________ degrees

10. Consider $(2\sin\theta - \sqrt{3})(2\sin\theta + \sqrt{3}) = 0$ on $0° \le \theta < 360°$.

(a) Find all solutions on the interval.

Answer: ________________ degrees

(b) The equation $2\sin\theta - \sqrt{3} = 0$ has only two solutions on the same interval. Explain why the equation in part (a) has four.

Answer: ________________